Derivations of Exceptions Prog Tests in FreeST

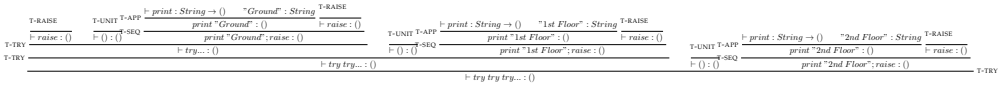
Mestrado em Engenharia Informática
Faculdade de Ciências da Universidade de Lisboa

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1 RaiseRaiseRaise

1.1 Type



2 TryReceiveChainCancel2 (Old)

2.1 Evaluation

$$
\begin{array}{ll}
\quad (va)(vb)(sendInts\ (1,2,4)\ a \mid\mid sumInts\ b) & \\
\rightarrow (va)(vb)(a \triangleright send\ 1 \triangleright cancel \mid\mid try\ receive\ b\ as\ (n1,b)\ in\ M\ otherwise\ 0 \mid\mid a(\epsilon) \rightsquigarrow b(\epsilon)) & by\ definition \\
\rightarrow (va)(vb)(a \triangleright cancel \mid\mid try\ receive\ b\ as\ (n1,b)\ in\ M\ otherwise\ 0 \mid\mid a(\epsilon) \rightsquigarrow b(1)) & by\ \mathtt{E-SEND} \\
\rightarrow (va)(vb)(() \mid\mid \lightning\ a \mid\mid try\ receive\ b\ as\ (n1,b)\ in\ M\ otherwise\ 0 \mid\mid a(\epsilon) \rightsquigarrow b(1)) & by\ \mathtt{E-CANCEL} \\
\rightarrow (va)(vb)(() \mid\mid \lightning\ a \mid\mid try\ (1,b)\ as\ (n1,b)\ in\ M\ otherwise\ 0 \mid\mid a(\epsilon) \rightsquigarrow b(\epsilon)) & by\ \mathtt{E-RECEIVE} \\
\rightarrow (va)(vb)(() \mid\mid \lightning\ a \mid\mid M[(1,b)/(n1,b)] \mid\mid a(\epsilon) \rightsquigarrow b(\epsilon)) & by\ \mathtt{E-VAL} \\
\rightarrow (va)(vb)(() \mid\mid \lightning\ a \mid\mid try\ receive\ b\ as\ (n2,b)\ in ...otherwise\ 1 \mid\mid a(\epsilon) \rightsquigarrow b(\epsilon)) & by\ definition\ of\ M \\
\rightarrow (va)(vb)(() \mid\mid \lightning\ a \mid\mid try\ raise\ as\ (n2,b)\ in ...otherwise\ 1 \mid\mid \lightning\ b \mid\mid a(\epsilon) \rightsquigarrow b(\epsilon)) & by\ \mathtt{E-RECEIVEZAP} \\
\rightarrow (va)(vb)(() \mid\mid \lightning\ a \mid\mid 1 \mid\mid \lightning\ b \mid\mid a(\epsilon) \rightsquigarrow b(\epsilon)) & by\ \mathtt{E-RAISE} \\
\equiv () \mid\mid 1 & \\
\equiv 1 &
\end{array}
$$

3 AllChannelsCancelled (Old)

3.1 Evaluation

$$
\begin{array}{ll}
\quad \mathit{let}\ (w, r) = \mathit{channel}\ @\mathit{Three}\ \mathit{in}\ \mathit{sendInts}\ (1,2,4)\ w;\ \mathit{print}\ \$\ \mathit{sumInts}\ r & \\
\to (va1)(vb1)(\mathit{let}\ (a1, b1) = (a1, b1)\ \mathit{in}\ \mathit{sendInts}\ (1,2,4)\ a1;\ \mathit{print}\ \$\ \mathit{sumInts}\ b1 \mid\mid a1(\epsilon) \leftrightsquigarrow b1(\epsilon)) & \mathit{by}\ \mathtt{E-LET}\ \mathit{and}\ \mathit{channel} \\
\to (va1)(vb1)(\mathit{sendInts}\ (1,2,4)\ a1;\ \mathit{print}\ \$\ \mathit{sumInts}\ b1 \mid\mid a1(\epsilon) \leftrightsquigarrow b1(\epsilon)) & \mathit{by}\ \mathtt{E-LETV} \\
\to (va1)(vb1)(\mathit{let}\ (a2, b2) = \mathit{channel}\ @\mathit{Two}\ \mathit{in}...;\ \mathit{print}\ \$\ \mathit{sumInts}\ b1 \mid\mid a1(\epsilon) \leftrightsquigarrow b1(\epsilon)) & \mathit{by\ definition\ of\ sendInts} \\
\to (va1)(vb1)(va2)(vb2)(\mathit{let}\ (a3, b3) = \mathit{channel}\ @\mathit{One}\ \mathit{in}...;\ \mathit{print}\ \$\ \mathit{sumInts}\ b1 \mid\mid a1(\epsilon) \leftrightsquigarrow b1(\epsilon) \mid\mid a2(\epsilon) \leftrightsquigarrow b2(\epsilon)) & \mathit{by}\ \mathtt{E-LET}\ \mathit{and}\ \mathtt{E-LETV} \\
\to (va1)(vb1)(va2)(vb2)(va3)(vb3)(\mathit{let}\ a1 = a1 \triangleright \mathit{send}\ 1 \triangleright \mathit{send}\ b2\ \mathit{in}...;\ \mathit{print}\ \$\ \mathit{sumInts}\ b1 & \\
\quad \mid\mid a1(\epsilon) \leftrightsquigarrow b1(\epsilon) \mid\mid a2(\epsilon) \leftrightsquigarrow b2(\epsilon) \mid\mid a3(\epsilon) \leftrightsquigarrow b3(\epsilon)) & \mathit{by}\ \mathtt{E-LET}\ \mathit{and}\ \mathtt{E-LETV} \\
\to (va1)(vb1)(va2)(vb2)(va3)(vb3)(\mathit{let}\ a2 = a2 \triangleright \mathit{send}\ 2 \triangleright \mathit{send}\ b3\ \mathit{in}...;\ \mathit{print}\ \$\ \mathit{sumInts}\ b1 & \\
\quad \mid\mid a1(\epsilon) \leftrightsquigarrow b1(1\ b2) \mid\mid a2(\epsilon) \leftrightsquigarrow b2(\epsilon) \mid\mid a3(\epsilon) \leftrightsquigarrow b3(\epsilon)) & \mathit{by}\ \mathtt{E-SEND},\ \mathtt{E-LET}\ \mathit{and}\ \mathtt{E-LETV} \\
\to (va1)(vb1)(va2)(vb2)(va3)(vb3)((\mathit{let}\ a3 = a3 \triangleright \mathit{send}\ 4\ \mathit{in}\ \mathit{close}\ a3;\ \mathit{close}\ a2;\ \mathit{close}\ a1);\ \mathit{print}\ \$\ \mathit{sumInts}\ b1 & \\
\quad \mid\mid a1(\epsilon) \leftrightsquigarrow b1(1\ b2) \mid\mid a2(\epsilon) \leftrightsquigarrow b2(2\ b3) \mid\mid a3(\epsilon) \leftrightsquigarrow b3(\epsilon)) & \mathit{by}\ \mathtt{E-SEND},\ \mathtt{E-LET}\ \mathit{and}\ \mathtt{E-LETV} \\
\to (va1)(vb1)(va2)(vb2)(va3)(vb3)((\mathit{close}\ a3;\ \mathit{close}\ a2;\ \mathit{close}\ a1);\ \mathit{print}\ \$\ \mathit{sumInts}\ b1 & \\
\quad \mid\mid a1(\epsilon) \leftrightsquigarrow b1(1\ b2) \mid\mid a2(\epsilon) \leftrightsquigarrow b2(2\ b3) \mid\mid a3(\epsilon) \leftrightsquigarrow b3(4)) & \mathit{by}\ \mathtt{E-SEND},\ \mathtt{E-LET}\ \mathit{and}\ \mathtt{E-LETV} \\
\to (va1)(vb1)(va2)(vb2)(va3)(vb3)(\mathit{print}\ \$\ \mathit{sumInts}\ b1 & \\
\quad \mid\mid a1(\epsilon) \leftrightsquigarrow b1(1\ b2\ \mathit{EOS}) \mid\mid a2(\epsilon) \leftrightsquigarrow b2(2\ b3\ \mathit{EOS}) \mid\mid a3(\epsilon) \leftrightsquigarrow b3(4\ \mathit{EOS})) & \mathit{by}\ \mathtt{E-CLOSE} \\
\to (va1)(vb1)(va2)(vb2)(va3)(vb3)(\mathit{print}\ \$\ \mathit{cancel}\ b1;\ \mathit{Nothing} & \\
\quad \mid\mid a1(\epsilon) \leftrightsquigarrow b1(1\ b2\ \mathit{EOS}) \mid\mid a2(\epsilon) \leftrightsquigarrow b2(2\ b3\ \mathit{EOS}) \mid\mid a3(\epsilon) \leftrightsquigarrow b3(4\ \mathit{EOS})) & \mathit{by\ definition\ of\ sumInts} \\
\to (va1)(vb1)(va2)(vb2)(va3)(vb3)(\mathit{print}\ \$\ ();\ \mathit{Nothing} \mid\mid \lightning\ b1 & \\
\quad \mid\mid a1(\epsilon) \leftrightsquigarrow b1(1\ b2\ \mathit{EOS}) \mid\mid a2(\epsilon) \leftrightsquigarrow b2(2\ b3\ \mathit{EOS}) \mid\mid a3(\epsilon) \leftrightsquigarrow b3(4\ \mathit{EOS})) & \mathit{by}\ \mathtt{E-CANCEL} \\
\to (va1)(vb1)(va2)(vb2)(va3)(vb3)(\mathit{print}\ \$\ ();\ \mathit{Nothing} \mid\mid \lightning\ b1 \mid\mid \lightning\ 1 \mid\mid \lightning\ b2 \mid\mid \lightning\ \mathit{EOS}(a1) & \\
\quad \mid\mid a1(\epsilon) \leftrightsquigarrow b1(\epsilon) \mid\mid a2(\epsilon) \leftrightsquigarrow b2(2\ b3\ \mathit{EOS}) \mid\mid a3(\epsilon) \leftrightsquigarrow b3(4\ \mathit{EOS})) & \mathit{by}\ 3x\mathtt{E-ZAP} \\
\to (va1)(vb1)(va2)(vb2)(va3)(vb3)(\mathit{print}\ \$\ ();\ \mathit{Nothing} \mid\mid \lightning\ b1 \mid\mid \lightning\ b2 \mid\mid \lightning\ a1 & \\
\quad \mid\mid a1(\epsilon) \leftrightsquigarrow b1(\epsilon) \mid\mid a2(\epsilon) \leftrightsquigarrow b2(2\ b3\ \mathit{EOS}) \mid\mid a3(\epsilon) \leftrightsquigarrow b3(4\ \mathit{EOS})) & \mathit{by}\ \mathtt{E-EOSZAP} \\
\equiv (va2)(vb2)(va3)(vb3)(\mathit{print}\ \$\ ();\ \mathit{Nothing} \mid\mid \lightning\ b2 & \\
\quad \mid\mid a2(\epsilon) \leftrightsquigarrow b2(2\ b3\ \mathit{EOS}) \mid\mid a3(\epsilon) \leftrightsquigarrow b3(4\ \mathit{EOS})) & \\
\to (va2)(vb2)(va3)(vb3)(\mathit{print}\ \$\ ();\ \mathit{Nothing} \mid\mid \lightning\ b2 \mid\mid \lightning\ 2 \mid\mid \lightning\ b3 \mid\mid \lightning\ \mathit{EOS}(a2) & \\
\quad \mid\mid a2(\epsilon) \leftrightsquigarrow b2(\epsilon) \mid\mid a3(\epsilon) \leftrightsquigarrow b3(4\ \mathit{EOS})) & \mathit{by}\ 3x\mathtt{E-ZAP} \\
\to (va2)(vb2)(va3)(vb3)(\mathit{print}\ \$\ ();\ \mathit{Nothing} \mid\mid \lightning\ b2 \mid\mid \lightning\ b3 \mid\mid \lightning\ a2 & \\
\quad \mid\mid a2(\epsilon) \leftrightsquigarrow b2(\epsilon) \mid\mid a3(\epsilon) \leftrightsquigarrow b3(4\ \mathit{EOS})) & \mathit{by}\ \mathtt{E-EOSZAP} \\
\equiv (va3)(vb3)(\mathit{print}\ \$\ ();\ \mathit{Nothing} \mid\mid \lightning\ b3 & \\
\quad \mid\mid a3(\epsilon) \leftrightsquigarrow b3(4\ \mathit{EOS})) & \\
\to (va3)(vb3)(\mathit{print}\ \$\ ();\ \mathit{Nothing} \mid\mid \lightning\ b3 \mid\mid \lightning\ 4 \mid\mid \lightning\ \mathit{EOS}(a3) & \\
\quad \mid\mid a3(\epsilon) \leftrightsquigarrow b3(\epsilon)) & \mathit{by}\ 3x\mathtt{E-ZAP} \\
\to (va3)(vb3)(\mathit{print}\ \$\ ();\ \mathit{Nothing} \mid\mid \lightning\ b3 \mid\mid \lightning\ a3 & \\
\quad \mid\mid a3(\epsilon) \leftrightsquigarrow b3(\epsilon)) & \mathit{by}\ \mathtt{E-EOSZAP} \\
\equiv \mathit{print}\ \$\ ();\ \mathit{Nothing} & \\
\to \mathit{print}\ \$\mathit{Nothing} & \mathit{by}\ \mathtt{E-SEQNEXT} \\
\to () & \mathit{by\ definition\ of\ print}
\end{array}
$$

4 AllChannelsCancelled

4.1 Evaluation

$$
\begin{array}{ll}
\quad \mathit{let}\ (w, r) = \mathit{channel}\ @\mathit{Three}\ \mathit{in}\ \mathit{sendInts}\ (1,2,4)\ w;\ \mathit{print}\ \$\ \mathit{sumInts}\ r & \\
\to (va1)(vb1)(\mathit{let}\ (a1, b1) = (a1, b1)\ \mathit{in}\ \mathit{sendInts}\ (1,2,4)\ a1;\ \mathit{print}\ \$\ \mathit{sumInts}\ b1 \parallel a1(\epsilon) \leftrightsquigarrow b1(\epsilon)) & \mathit{by}\ \mathtt{E-LET}\ \mathit{and}\ \mathit{channel} \\
\to (va1)(vb1)(\mathit{sendInts}\ (1,2,4)\ a1;\ \mathit{print}\ \$\ \mathit{sumInts}\ b1 \parallel a1(\epsilon) \leftrightsquigarrow b1(\epsilon)) & \mathit{by}\ \mathtt{E-LETV} \\
\to (va1)(vb1)(\mathit{let}\ (a2, b2) = \mathit{channel}\ @\mathit{Two}\ \mathit{in}...;\ \mathit{print}\ \$\ \mathit{sumInts}\ b1 \parallel a1(\epsilon) \leftrightsquigarrow b1(\epsilon)) & \mathit{by\ definition\ of\ sendInts} \\
\to (va1)(vb1)(va2)(vb2)(\mathit{let}\ (a3, b3) = \mathit{channel}\ @\mathit{One}\ \mathit{in}...;\ \mathit{print}\ \$\ \mathit{sumInts}\ b1 \parallel a1(\epsilon) \leftrightsquigarrow b1(\epsilon) \parallel a2(\epsilon) \leftrightsquigarrow b2(\epsilon)) & \mathit{by}\ \mathtt{E-LET}\ \mathit{and}\ \mathtt{E-LETV} \\
\to (va1)(vb1)(va2)(vb2)(va3)(vb3)(\mathit{let}\ a1 = a1 \triangleright \mathit{send}\ 1 \triangleright \mathit{send}\ b2\ \mathit{in}...;\ \mathit{print}\ \$\ \mathit{sumInts}\ b1 & \\
\quad \parallel a1(\epsilon) \leftrightsquigarrow b1(\epsilon) \parallel a2(\epsilon) \leftrightsquigarrow b2(\epsilon) \parallel a3(\epsilon) \leftrightsquigarrow b3(\epsilon)) & \mathit{by}\ \mathtt{E-LET}\ \mathit{and}\ \mathtt{E-LETV} \\
\to (va1)(vb1)(va2)(vb2)(va3)(vb3)(\mathit{let}\ a2 = a2 \triangleright \mathit{send}\ 2 \triangleright \mathit{send}\ b3\ \mathit{in}...;\ \mathit{print}\ \$\ \mathit{sumInts}\ b1 & \\
\quad \parallel a1(\epsilon) \leftrightsquigarrow b1(1\ b2) \parallel a2(\epsilon) \leftrightsquigarrow b2(\epsilon) \parallel a3(\epsilon) \leftrightsquigarrow b3(\epsilon)) & \mathit{by}\ \mathtt{E-SEND},\ \mathtt{E-LET}\ \mathit{and}\ \mathtt{E-LETV} \\
\to (va1)(vb1)(va2)(vb2)(va3)(vb3)((\mathit{let}\ a3 = a3 \triangleright \mathit{send}\ 4\ \mathit{in}\ \mathit{close}\ a3;\ \mathit{close}\ a2;\ \mathit{close}\ a1);\ \mathit{print}\ \$\ \mathit{sumInts}\ b1 & \\
\quad \parallel a1(\epsilon) \leftrightsquigarrow b1(1\ b2) \parallel a2(\epsilon) \leftrightsquigarrow b2(2\ b3) \parallel a3(\epsilon) \leftrightsquigarrow b3(\epsilon)) & \mathit{by}\ \mathtt{E-SEND},\ \mathtt{E-LET}\ \mathit{and}\ \mathtt{E-LETV} \\
\to (va1)(vb1)(va2)(vb2)(va3)(vb3)((\mathit{close}\ a3;\ \mathit{close}\ a2;\ \mathit{close}\ a1);\ \mathit{print}\ \$\ \mathit{sumInts}\ b1 & \\
\quad \parallel a1(\epsilon) \leftrightsquigarrow b1(1\ b2) \parallel a2(\epsilon) \leftrightsquigarrow b2(2\ b3) \parallel a3(\epsilon) \leftrightsquigarrow b3(4)) & \mathit{by}\ \mathtt{E-SEND},\ \mathtt{E-LET}\ \mathit{and}\ \mathtt{E-LETV} \\
\to (va1)(vb1)(va2)(vb2)(va3)(vb3)(\mathit{print}\ \$\ \mathit{sumInts}\ b1 \parallel \lightning a1 \parallel \lightning a2 \parallel \lightning a3 & \\
\quad \parallel a1(\epsilon) \leftrightsquigarrow b1(1\ b2\ \mathit{EOS}) \parallel a2(\epsilon) \leftrightsquigarrow b2(2\ b3\ \mathit{EOS}) \parallel a3(\epsilon) \leftrightsquigarrow b3(4\ \mathit{EOS})) & \mathit{by}\ \mathtt{E-CLOSE} \\
\to (va1)(vb1)(va2)(vb2)(va3)(vb3)(\mathit{print}\ \$\ \mathit{cancel}\ b1;\ \mathit{Nothing} \parallel \lightning a1 \parallel \lightning a2 \parallel \lightning a3 & \\
\quad \parallel a1(\epsilon) \leftrightsquigarrow b1(1\ b2\ \mathit{EOS}) \parallel a2(\epsilon) \leftrightsquigarrow b2(2\ b3\ \mathit{EOS}) \parallel a3(\epsilon) \leftrightsquigarrow b3(4\ \mathit{EOS})) & \mathit{by\ definition\ of\ sumInts} \\
\to (va1)(vb1)(va2)(vb2)(va3)(vb3)(\mathit{print}\ \$\ ();\ \mathit{Nothing} \parallel \lightning a1 \parallel \lightning a2 \parallel \lightning a3 \parallel \lightning b1 & \\
\quad \parallel a1(\epsilon) \leftrightsquigarrow b1(1\ b2\ \mathit{EOS}) \parallel a2(\epsilon) \leftrightsquigarrow b2(2\ b3\ \mathit{EOS}) \parallel a3(\epsilon) \leftrightsquigarrow b3(4\ \mathit{EOS})) & \mathit{by}\ \mathtt{E-CANCEL} \\
\to (va1)(vb1)(va2)(vb2)(va3)(vb3)(\mathit{print}\ \$\ ();\ \mathit{Nothing} \parallel \lightning a1 \parallel \lightning a2 \parallel \lightning a3 \parallel \lightning b1 \parallel \lightning 1 \parallel \lightning b2 \parallel \lightning \mathit{EOS} & \\
\quad \parallel a1(\epsilon) \leftrightsquigarrow b1(\epsilon) \parallel a2(\epsilon) \leftrightsquigarrow b2(2\ b3\ \mathit{EOS}) \parallel a3(\epsilon) \leftrightsquigarrow b3(4\ \mathit{EOS})) & \mathit{by}\ 3x\mathtt{E-ZAP} \\
\equiv (va2)(vb2)(va3)(vb3)(\mathit{print}\ \$\ ();\ \mathit{Nothing} \parallel \lightning a2 \parallel \lightning a3 \parallel \lightning b2 & \\
\quad \parallel a2(\epsilon) \leftrightsquigarrow b2(2\ b3\ \mathit{EOS}) \parallel a3(\epsilon) \leftrightsquigarrow b3(4\ \mathit{EOS})) & \\
\to (va2)(vb2)(va3)(vb3)(\mathit{print}\ \$\ ();\ \mathit{Nothing} \parallel \lightning a2 \parallel \lightning a3 \parallel \lightning b2 \parallel \lightning 2 \parallel \lightning b3 \parallel \lightning \mathit{EOS} & \\
\quad \parallel a2(\epsilon) \leftrightsquigarrow b2(\epsilon) \parallel a3(\epsilon) \leftrightsquigarrow b3(4\ \mathit{EOS})) & \mathit{by}\ 3x\mathtt{E-ZAP} \\
\equiv (va3)(vb3)(\mathit{print}\ \$\ ();\ \mathit{Nothing} \parallel \lightning a3 \parallel \lightning b3 & \\
\quad \parallel a3(\epsilon) \leftrightsquigarrow b3(4\ \mathit{EOS})) & \\
\to (va3)(vb3)(\mathit{print}\ \$\ ();\ \mathit{Nothing} \parallel \lightning a3 \parallel \lightning b3 \parallel \lightning 4 \parallel \lightning \mathit{EOS} & \\
\quad \parallel a3(\epsilon) \leftrightsquigarrow b3(\epsilon)) & \mathit{by}\ 2x\mathtt{E-ZAP} \\
\equiv \mathit{print}\ \$\ ();\ \mathit{Nothing} & \\
\to \mathit{print}\ \$\mathit{Nothing} & \mathit{by}\ \mathtt{E-SEQNEXT} \\
\to () & \mathit{by\ definition\ of\ print}
\end{array}
$$